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Financial Impact Engine

A major challenge in airline operations is that the true cost of a delay is often invisible until it is too late. While dispatchers can see that a flight is running behind schedule, they rarely have a direct estimate of the financial consequences associated with that delay.

To address this, we developed a Financial Impact Engine that translates operational disruptions into estimated monetary losses.

The framework combines three sources of cost:

1. **Variable operating costs**, such as additional crew time, fuel burn, maintenance accrual and airport resource usage.
2. **Reactionary costs**, caused by delay propagation through the network, missed connections and downstream disruptions.
3. **Passenger compensation and care obligations**, based on EU Regulation 261/2004.

Because delay costs do not increase linearly, we model them using piecewise cost functions. Flights are first classified into aircraft-size categories using route distance as a proxy:

Size	Distance
Small	< 1,500 km
Medium	1,500–5,000 km
Large	> 5,000 km

Each category is assigned a representative passenger count and cost structure. The predicted delay duration is then mapped into operational severity brackets that reflect how costs escalate over time.

The framework captures several critical thresholds, including:

- Schedule buffer exhaustion
- Delay propagation to subsequent flights
- Passenger meal and refreshment obligations
- EU261 compensation thresholds
- Cancellation economics

As a result, the system can estimate the expected financial exposure associated with a predicted delay and communicate it directly to dispatchers in euros rather than probabilities alone.

Business Value Proposition

Most delay prediction tools answer the question:

> "Will this flight be delayed?"

Wingman answers the question:

> "How much will this delay cost us if we do nothing?"

This distinction fundamentally changes how operational decisions are made.

A delay with a high probability is not necessarily the most important flight in the network. Conversely, a moderately risky flight carrying significant compensation exposure may represent a much larger financial threat to the airline.

By estimating the expected cost of disruption, Wingman enables dispatchers to prioritise flights based on business impact rather than probability alone.

This creates a direct connection between operational decisions and financial outcomes. Instead of reacting to delays after they occur, controllers can focus their attention on the flights where intervention is expected to prevent the greatest economic loss.

For airlines, this means:

- Reduced compensation payouts
- Reduced network disruption
- Better utilisation of operational resources
- Faster prioritisation during irregular operations
- Improved decision consistency across shift handovers

Wingman transforms disruption management from a reactive process into a proactive value-protection

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Unique Selling Point

The unique innovation behind Wingman is its ability to convert operational risk into financial exposure using only information available before departure.

Using route distance, delay duration, and aircraft-size classification, the system estimates the economic impact of a disruption without requiring access to proprietary airline accounting systems.

For every flight, Wingman can:

- Predict disruption risk
- Explain the likely cause of the disruption
- Estimate the financial impact in euros
- Classify the disruption into operational severity brackets
- Prioritise flights according to expected avoidable loss

This allows dispatchers to move beyond simple risk scores and understand the real business consequences of inaction.

A flight that is predicted to be delayed by 90 minutes is no longer just a red warning on a dashboard. It becomes a quantified business event with a measurable financial impact.

By making delay costs visible before they occur, Wingman provides a layer of decision intelligence that is largely absent from today's operations-control tooling.